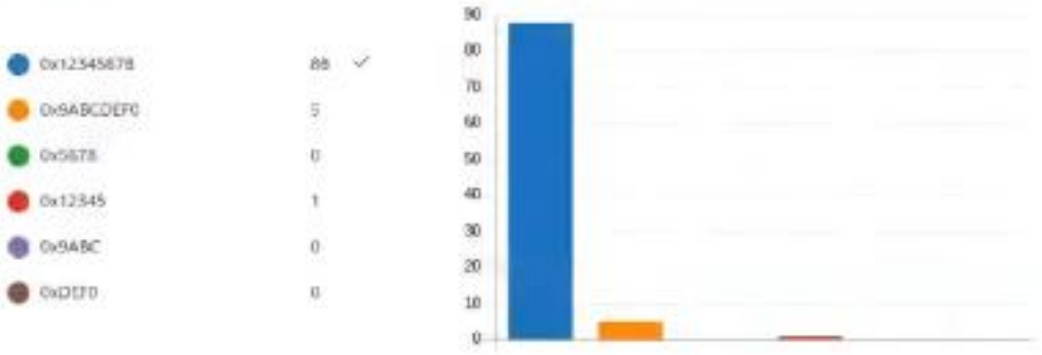
From 5A to 9B Quizzzzzzzzzzzzzzzzzzzzzzzzzzzzzz

1. Let's assume we defined an array using m: .word 0x12345678 .word 0x9ABCDEF0 What would be the content of r1 register if we do LDR r0, =m LDR r1, [r0] (20 points)

94% of respondents (88 of 94) answered this question correctly.

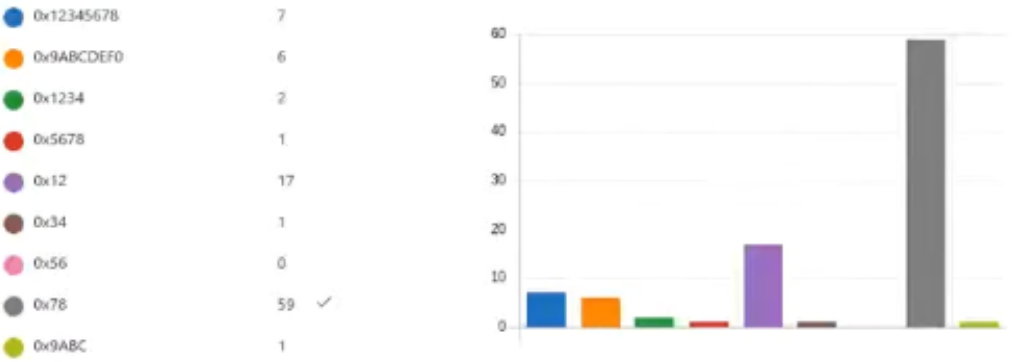
More Details



2. Let's assume we defined an array using m: .word 0x12345678 .word 0x9ABCDEF0 What would be the content of r1 register if we do LDR r0, =m LDRB r1, [r0] (20 points)

63% of respondents (59 of 94) answered this question correctly.

More Details



3. Which of the following is INCORRECT? (20 points)

96% of respondents (90 of 94) answered this question correctly.

More Details



4. Which of the following is correct? (20 points)

90% of respondents (85 of 94) answered this question correctly.

More Details



5. Which one of the following is NOT a possible reason for HardFault on Cortex-M0+? **(20 points)**

88% of respondents (83 of 94) answered this question correctly.

More Details

Divide by zero	83	✓
Access invalid address	3	
Access unaligned memory	3	
Undefined instruction	5	



1. Which one of the following is the most portable (and correct) way for declaring a channel frequency of 14 Ghz?
(10 Points)

- #define CHFREQ 1000 * 1000 * 1000 * 14
- #define CHFREQ 1000 * 1000 * 1000 * 14UL
- #define CHFREQ (1000 * 1000 * 1000 * 14)
- #define CHFREQ (1000 * 1000 * 1000 * 14UL) ✓
- #define CHFREQ 1000 * 1000 * 1000 * 14;

2. Which is the most flexible and error-free way to declare a C preprocessor macro that will find the smaller of the two given values?
(10 Points)

- #define MIN(A, B) ((A) < (B) ? (A) : (B)) ✓
- #define MIN(A, B) { if (A < B) A; else B;}
- #define MIN(A, B) (A < B ? A : B)
- #define MIN(A, B) A < B ? A : B;

3. Which of the wollowing will create an infinite loop in C?
(10 Points)

- while(1) { ... }
- for(;;) { ... }
- loop: ... goto loop;
- All of them ✓

4. Which is true for static keyword for C? **(10 points)**

72% of respondents (69 of 96) answered this question correctly.

More Details

A static variable declared withi...	14	
A static variable declared outsi...	7	
Both statements are correct	69	✓
Both statements are INCORRE...	6	

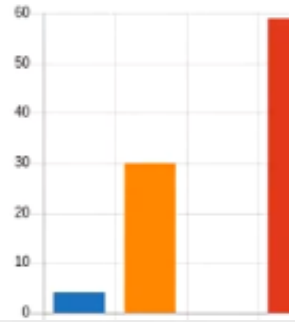


5. Which can be used to reset LSB of A? (10 points)

61% of respondents (59 of 96) answered this question correctly.

More Details

A & 0x01;	4
A & ~0x01;	30
A | ~0x01;	0
A &= ~0x01;	59 ✓
A |= 0x01;	3
A |= ~0x01;	0
A |= 0x80;	0



6. Which is the correct way of writing value 0x1234 to memory address 0x8000 (10 Points)

- uint16 *p = (uint16_t *) 0x8000; p = 0x1234;
- uint16 *p = (uint16_t *) 0x1234; p = 0x8000;
- * (uint16 * const)(0x8000) = 0x1234; ✓
- * (uint16 * const)(0x1234) = 0x8000;

7. Which is the correct place to declare variables with volatile keyword in C? (10 Points)

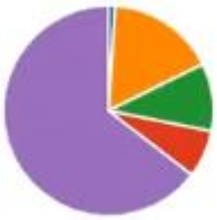
- Memory-mapped IO
- Global variables that are accessible by multiple threads in multi-threaded environments
- Global variables accessible by any interrupt service routines
- All of them ✓

8. When this code is compiled, what section will X go? int hede() { int x = 0; ... } (10 points)

65% of respondents (62 of 96) answered this question correctly.

More Details

text	1
data	16
bss	10
heap	7
stack	62 ✓

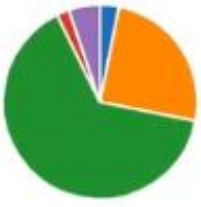


9. When this code is compiled, what section will X go? int hede() { static int x = 0; ... } (10 points)

65% of respondents (62 of 96) answered this question correctly.

More Details

text	3
data	24
bss	62 ✓
heap	2
stack	5

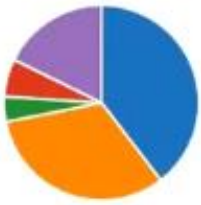


10. When this code is compiled, what section will X go? const int x = 3; int hede() { ... } (10 points)

40% of respondents (38 of 96) answered this question correctly.

More Details

text	38	✓
data	31	
bss	4	
heap	6	
stack	17	

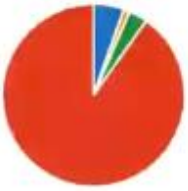


1. Which of the following is not a common fault injection techniques (10 points)

90% of respondents (83 of 92) answered this question correctly.

More Details

Glitch attack	5	
Temperature attack	1	
Magnetic attack	3	
Verbal attack	83	✓

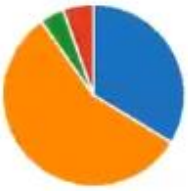


2. An interrupt service routine is usually a (10 points)

57% of respondents (52 of 92) answered this question correctly.

More Details

hardware-triggered synchrono...	31	
hardware-triggered asynchron...	52	✓
software-triggered asynchron...	4	
software-triggered synchrono...	5	



3. Which of the following stpes is not part of the CPU's hardwired exception processing? (10 points)

79% of respondents (72 of 91) answered this question correctly.

More Details

Allocate a heap for data storage	72	✓
Push context onto current stack	10	
Switch to handler/privileged ...	4	
Start executing code of interru...	5	

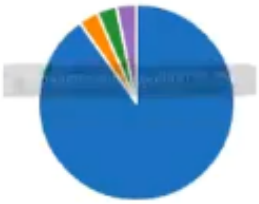


4. Which of the following is not a fault injection method? (10 points)

90% of respondents (83 of 92) answered this question correctly.

More Details

Interrupt Fault Injection	83	✓
Clock Fault Injection	3	
Optical Fault Injection	3	
Electromagnetic Fault Injection	0	
Voltage Fault Injection	3	



5. When an interruption is requested, which of the following registers is not pushed onto the current stack? (10 points)

41% of respondents (38 of 92) answered this question correctly.

More Details

R0	4	
R15	43	
R14	7	
R5	38	✓

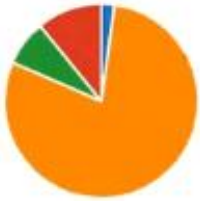


6. Which of the following registers is loaded with the EXC_RETURN code when an interrupt is triggered? (10 points)

79% of respondents (73 of 92) answered this question correctly.

More Details

R13	2	
R14	73	✓
R15	7	
None of them	10	



7. What type of interrupts are exceptions, faults and software interrupts? (10 points)

40% of respondents (37 of 92) answered this question correctly.

More Details

asynchronous	55	
synchronous	37	✓



8. Which of the following is not an exception state of an NVIC? (10 points)

76% of respondents (70 of 92) answered this question correctly.

More Details

Inactive	3	
Active and Pending	17	
Pending and Inactive	70	✓
Active	2	



9. Exceptions are prioritized to order the response of simultaneous requests. Which of the following is true? (10 points)

87% of respondents (80 of 92) answered this question correctly.

More Details

Smaller number = lower priority	12	
Smaller number = higher prior...	80	✓



10. How can race conditions be avoided? (10 points)

50% of respondents (46 of 92) answered this question correctly.

More Details

Declare used variables in critic...	46	
Prevent preemption within crit...	46	✓



1. General-purpose input and output (GPIO) can be used as (10 points)

92% of respondents (78 of 85) answered this question correctly.

More Details

Analog input	0
Analog Output	0
Digital I/O	7
All of them	78 ✓



2. Which one of the following is a correct way to connect an LED to MC GPIO pin? (10 points)

35% of respondents (30 of 85) answered this question correctly.

More Details

with a series resistor to VDD	30 ✓
with a pull-up resistor to VDD	4
with a pull-down resistor to G...	51
directly to VDD	0
directly to GND	0

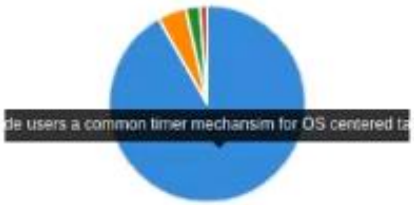


3. What is the purpose of the SysTick Timer? (10 points)

92% of respondents (78 of 85) answered this question correctly.

More Details

To provide users a common ti...	78 ✓
To provide users another timer	4
To detect faults and take coun...	2
To detect external events	1

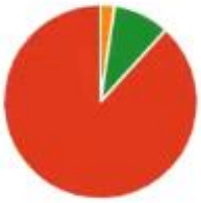


4. What is the purpose of the External Interrupts? (10 points)

88% of respondents (75 of 85) answered this question correctly.

More Details

To provide users a common ti...	0
To provide users another timer	2
To detect faults and take coun...	8
To detect external events	75 ✓

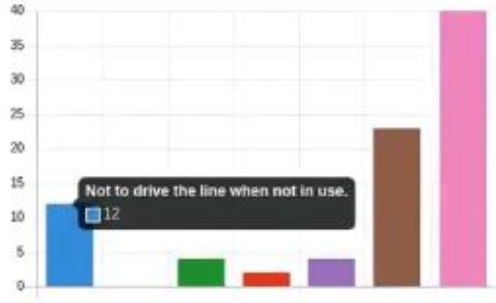


5. In the GPIO peripheral, what is the purpose of the open-drain pin configuration (or circuitry)? (10 points)

14% of respondents (12 of 85) answered this question correctly.

More Details

Not to drive the line when not...	12 ✓
Filter out unwanted noise with...	0
Filter out unwanted noise with...	4
Filter out unwanted noise with...	2
To act as an input to detect th...	4
To serve as a pull up	23
To serve as a pull down	40

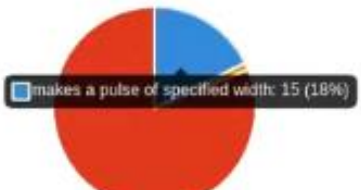


6. The PWM Module **(10 points)**

18% of respondents (15 of 85) answered this question correctly.

More Details

makes a pulse of specified wid...	15	✓
makes a pulse after specified ...	1	
toggle output signal pulse	0	
makes a series of pulses of sp...	69	



7. The duty cycle of PWM signal can be calculated as **(10 points)**

92% of respondents (78 of 85) answered this question correctly.

More Details

Output voltage / power	2	
Kirchhoff	1	
Frequency / period	4	
on-time / period	78	✓
on-time / off-time	0	



8. When connected to an external signal, the Timer peripheral can be used as **(10 points)**

48% of respondents (41 of 85) answered this question correctly.

More Details

to calculate the date	1	
to count external events	41	✓
respond quickly to interrupts	42	
reload I/O ports after a reset	1	



9. The reading of input signals is synchronized with the MCU clock **(10 points)**

81% of respondents (69 of 85) answered this question correctly.

More Details

True	69	✓
False	16	



10. The GPIO input circuitry needs to accurately and reliably recognize logic 1 and logic 0 signals. It does this by **(10 points)**

63% of respondents (53 of 84) answered this question correctly.

More Details

Precisely defining the input vo...	10	
Using an analog voltage comp...	18	
Blocking alternating signals to...	3	
Defining ranges of voltages fo...	53	✓

